Now this means that there exists a way to *measure x, y,* and z on three perpendicular axes, and the forces along those directions, such that these laws are true. These must be measured from some origin, but *where do we put the origin*? All that Newton would tell us at first is that there *is* some place that we can measure from, perhaps the center of the universe, such that these laws are correct. But we can show immediately that we can never find the center, because if we use some other origin it would make no difference. In other words, suppose that there are two people—Joe, who has an origin in one place, and Moe, who has a parallel system whose origin is somewhere else (Figure 1-1). Now when Joe measures the location of the point in space, he finds it at x, y, and z (we shall usually leave z out because it is too confusing to draw in a picture). Moe, on the other hand, when measuring the same point, will obtain a different x (in order to distinguish it, we will call it x'), and in principle a different y, although in our example they are numerically equal. So we have

$$x' = x - a, \quad y' = y, \quad z' = z. \tag{1.2}$$

Now in order to complete our analysis we must know what Moe would obtain for the forces. The force is supposed to act along some line, and by the force in the x-direction we mean the part of the total which is in the x-direction, which is the magnitude of the force times this cosine of its angle with the x-axis. Now we see that

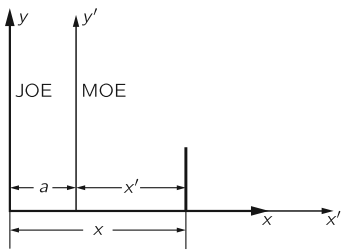

Figure 1-1 Two parallel coordinate systems.